

ENCN205

Applied Data Analysis for Civil and Natural Systems

LECTURE 8

Hydrological Analysis



Slides and interactive demos (click or scan):
aardid.github.io/uc_205_gis_course

University of Canterbury | Te Whare Wānanga o Waitaha

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Water flows downhill -- GIS can trace every drop.

Where does rainwater go when it hits the ground?

Which streams feed a reservoir? How large is the catchment?

If we build on this hillslope, where does stormwater drain?

How does urbanisation change the runoff pattern?

A DEM + flow algorithms answer all of these questions.

Why DEM-Based Hydrology for Engineering?

Water engineering starts with understanding the catchment:

Flood risk assessment — how much water can a catchment deliver?

Stormwater design — sizing pipes and detention ponds

Dam and reservoir planning — catchment area determines yield

Erosion and sediment modelling — flow paths determine where soil moves

Environmental flows — how much water reaches rivers and wetlands?

Before GIS: manual catchment delineation on topo maps — hours of work per catchment.

With GIS: automated, reproducible, scalable — thousands of catchments in minutes.

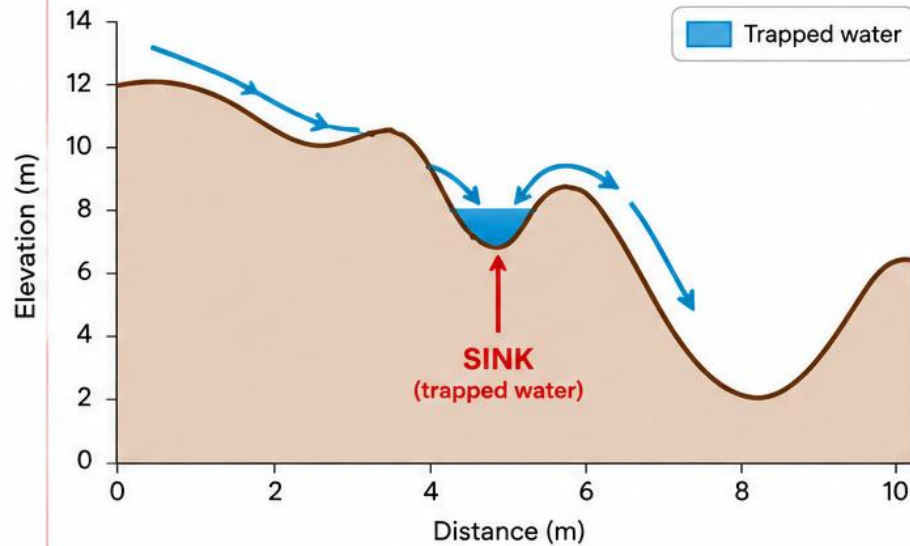
DEM Preprocessing — Sink Filling

Removing spurious depressions so water can flow continuously downhill.



1 BEFORE: DEM with Sink (Spurious Depression)

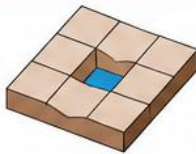
⚠ Water gets trapped — flow path is broken.



DEM cells (example)
(values in meters)

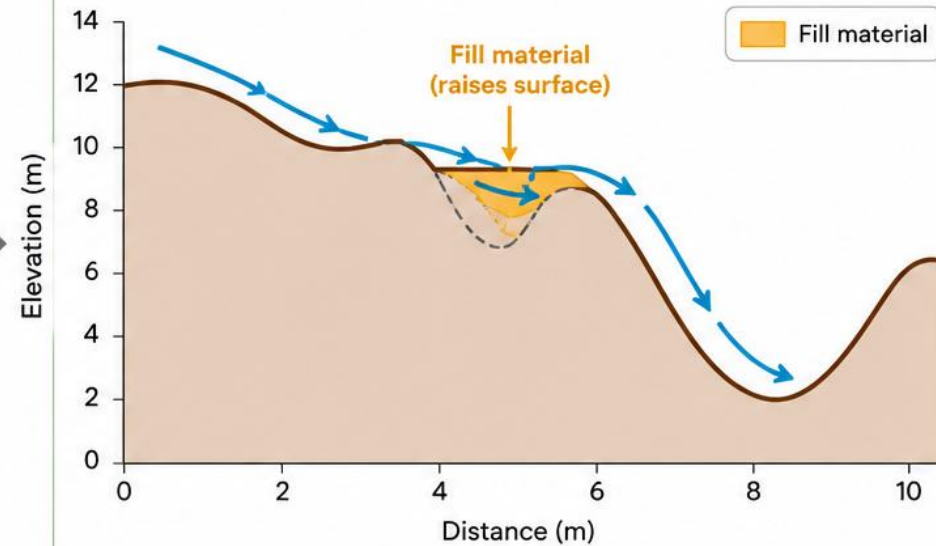
10	10	10
10	6	10
10	10	10

Low cell creates
a depression



2 AFTER: Filled DEM (Continuous Drainage)

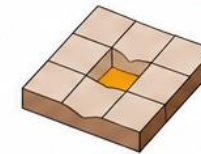
✓ Depression is filled — water flows downhill.



DEM cells (example)
(values in meters)

10	10	10
10	9	10
10	10	10

Low cell is raised
to eliminate
the depression



What is a sink?

A depression or low cell
surrounded by higher cells
that traps water.



Why fill sinks?

- Ensures continuous downhill flow
- Required for accurate hydrologic analysis (flow direction, accumulation, watersheds, stream networks)



How it works

Sinks are identified and their
elevation is raised just enough
to spill water out of the basin.



Key takeaway

Sink filling removes artefacts
in the DEM so water can flow
naturally across the landscape.

Note: True natural lakes/wetlands should be preserved if they are real features.

Always ask: 'was the DEM conditioned?'

Sinks trap water and break flow routing — they must be filled before analysis. Most sinks are DEM artefacts, not real depressions (but some are — e.g. volcanic craters).

D8 Flow Direction Algorithm

Each cell stores the direction of steepest descent to one of its 8 neighbours.

1 D8 Direction Encoding

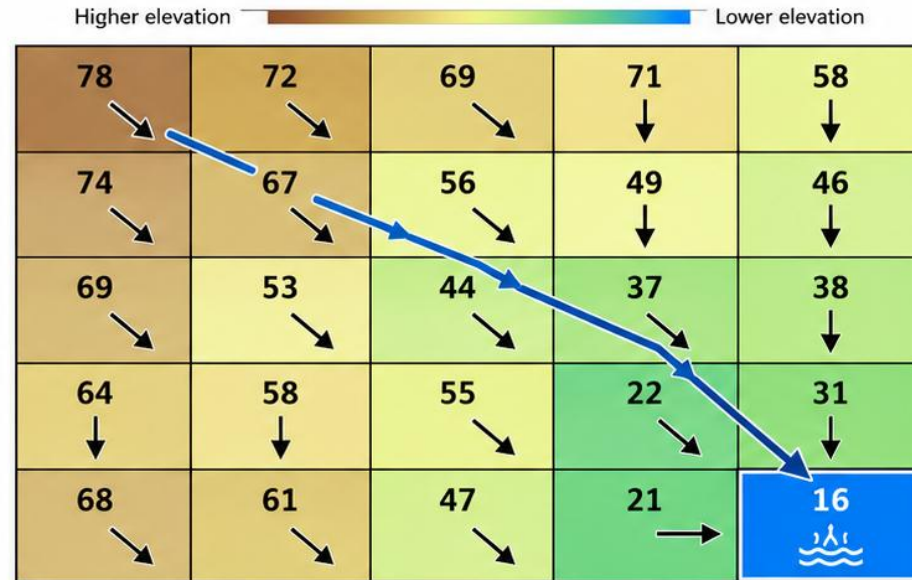
The number in each cell is the D8 code (power of 2) for flow to that neighbour.

128 NW ↖	64 N ↑	32 NE ↗
16 W ←	CENTER CELL (value)	1 E →
8 SW ↙	4 S ↓	2 SE ↘

i Each cell stores **ONE** flow direction — toward the neighbour with the steepest downward slope.

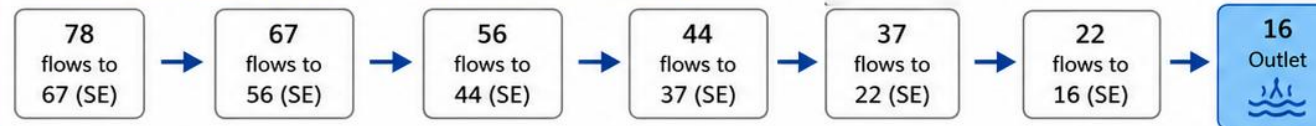
💡 D8 assumes water flows to **ONE** neighbouring cell only.

2 Example: Elevation Grid and D8 Flow Directions



← Catchment outlet

How it works (example path from top-left cell 78 to outlet)



✓ D8 routes all water from each cell to the single neighbour with the greatest downward slope.

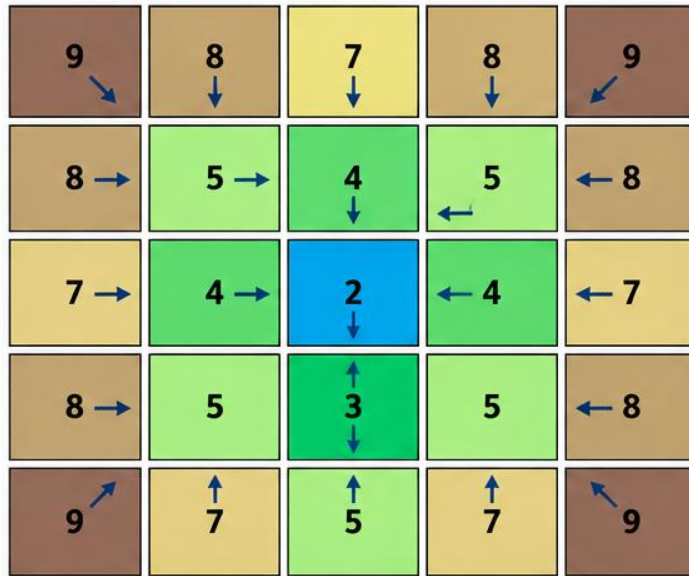
The 8 possible directions are encoded as powers of 2:
1, 2, 4, 8, 16, 32, 64, 128 (E, SE, S, SW, W, NW, N, NE)

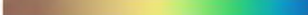
If all neighbours are higher, the cell is a sink (no outflow) unless pre-filled.

Flow Accumulation — Counting Upstream Cells

Flow accumulation counts how many cells drain to each cell (including itself).

1. Elevation + Flow Directions (D8)



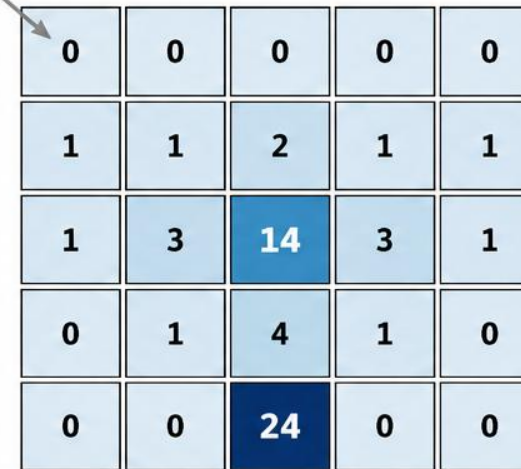
Elevation
High  Low

→ Flow direction
(steepest descent)

2. Flow Accumulation Count

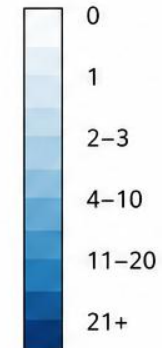
Number of upstream cells that drain to each cell (including itself).

Ridge
(0 upstream)



Valley / Outlet
(24 upstream)

Accumulation
(upstream cells)



What it shows

Brighter / darker blue cells receive flow from more upstream cells.

How it works

- Each cell starts with a count of 1 (itself).
- Cells pass their count to the downslope neighbour.
- Counts accumulate downstream.

Why it matters

- Identifies streams and drainage networks
- Highlights valleys and watershed structure
- Basis for stream ordering, delineation, and many hydrologic analyses

Key idea

Flow accumulation reveals where water converges in the landscape.



Note: This example uses the D8 flow direction algorithm (flow to one neighbour only).

High accumulation = valleys, streams, rivers. Zero = ridges and hilltops. Stream extraction: apply a threshold (e.g. accumulation > 100 cells = stream).

Stream Network Extraction

From flow accumulation to stream network:

1. Compute flow accumulation from flow direction grid
2. Choose an accumulation threshold (e.g. 500, 1000, 5000 cells)
3. Cells above threshold = stream network

Threshold choice affects network density:

Low threshold (100) = dense network with small tributaries

High threshold (5000) = only major rivers

`stream_mask = flow_accum > threshold`

Validate against known river maps (LINZ river centrelines, NZ River Environment Classification).

Always ask: 'what threshold defines a stream?'

LIVE DEMO

[L08_FlowAnalysis.html](#)

Adjust the accumulation threshold interactively to see how it controls stream network density and catchment size.

Scan to open the demo on your phone



[aardid.github.io/uc_205_gis_course/L08_FlowAnalysis.html](#)
All slides & demos: [aardid.github.io/uc_205_gis_course](#)

Catchment Delineation

Define a pour point (outlet) and trace all cells that drain to it:

Pour point: the location where you want to know the upstream area

Examples: dam site, bridge, stream gauge, river confluence

Algorithm:

1. Place pour point on the stream network (snap to high accumulation cell)
2. Trace upstream: follow flow direction arrows backwards
3. All traced cells = the catchment (watershed)

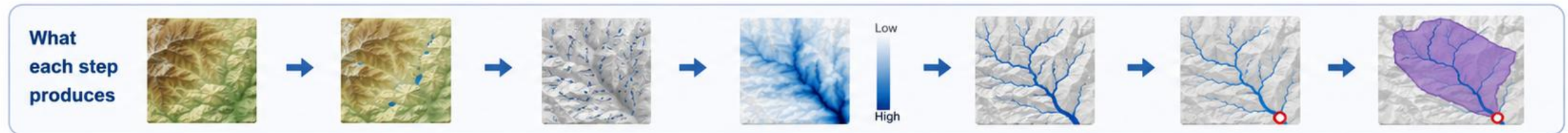
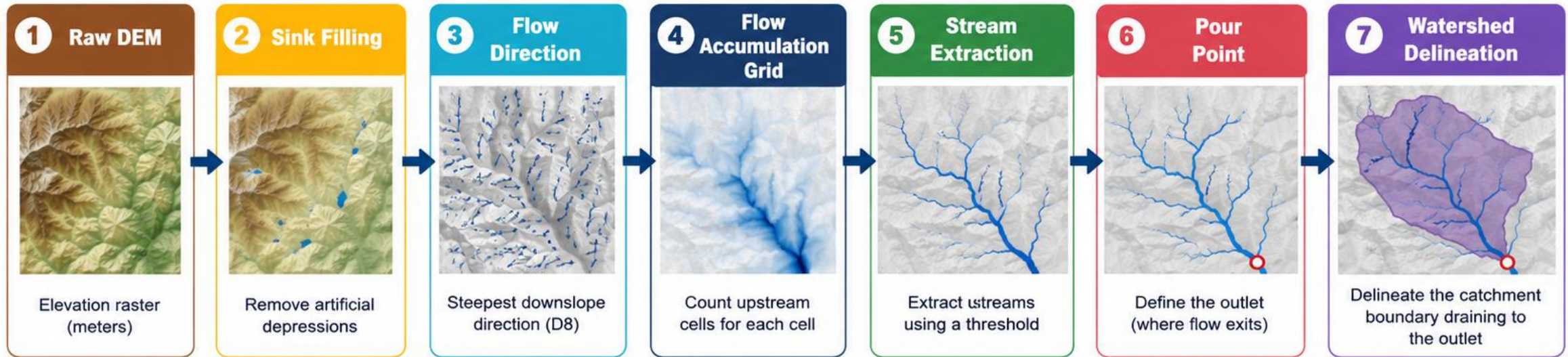
Critical step: snap pour point to the stream network.

If the pour point is off the network by even one cell, you get the wrong catchment.

Catchment Delineation Workflow



From raw elevation data to the watershed draining to a chosen outlet



Why this workflow?

Transforms elevation data into hydrologic information to understand drainage patterns and contributing areas.



Key outputs

- Flow direction grid
- Flow accumulation grid
- Stream network
- Watershed boundary



Common uses

- Flood modeling
- Water resource planning
- Erosion and sediment studies
- Watershed management



Note

Results depend on DEM quality, resolution, and parameter choices (e.g., stream threshold).



Blue arrows show flow direction. Water always flows from higher elevation to lower elevation.

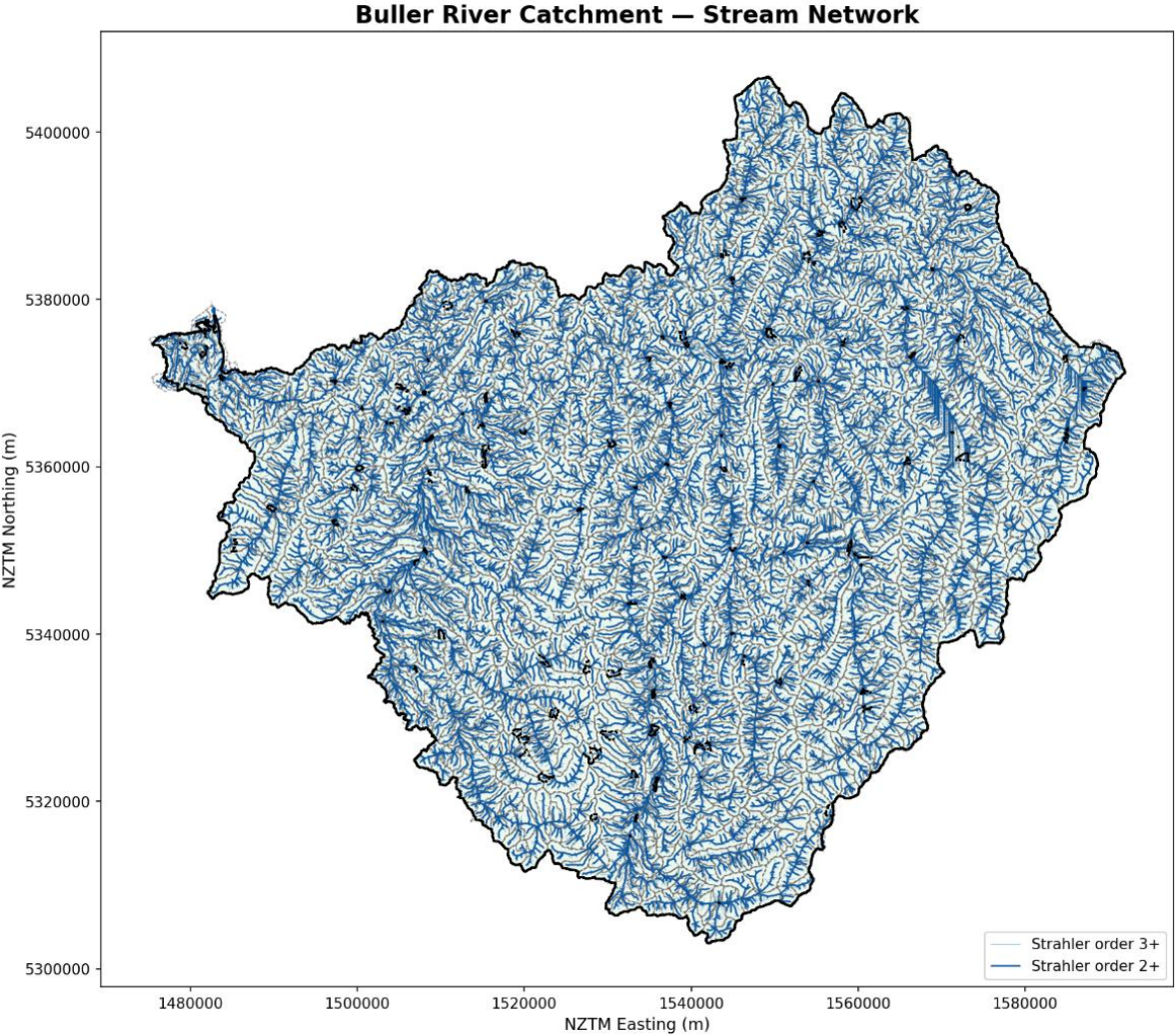
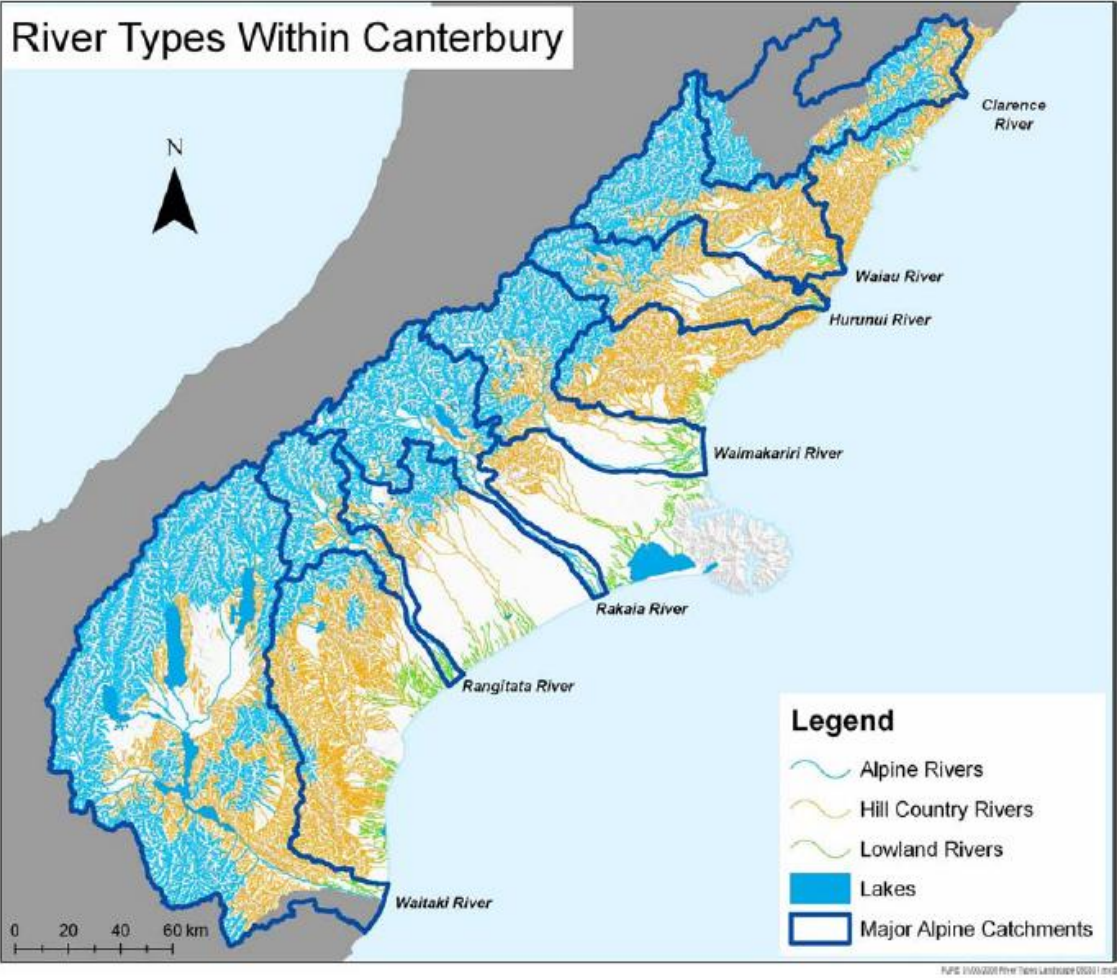


The pour point (outlet) defines where the catchment drains out of the system.

Seven steps from raw DEM to catchment boundary: Each step depends on the previous — errors propagate forward.

Most common error: skipping sink filling — leads to disconnected drainage.

River Catchments

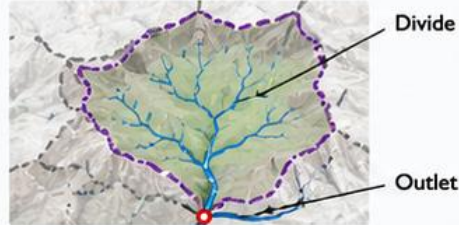


Watershed (Catchment) Characteristics

A watershed is the area of land that collects precipitation and drains to a common outlet.

1. Watershed Boundary

The divide (ridge line) separates one basin from another.



2. Stream Network

Streams form a branching network that conveys water to the outlet.



3. Drainage Area

The total area that contributes runoff to the outlet.

Drainage Area
= 38.6 km²

Divide (Ridge Line)

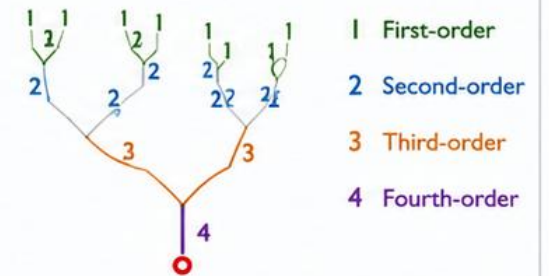


Key idea

All water that falls inside the boundary flows through the stream network to the outlet.

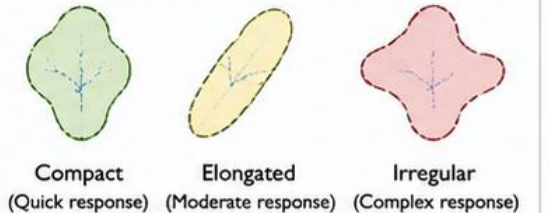
4. Basin Order (Strahler)

Streams are ranked by their position in the network.



5. Basin Shape

Shape influences hydrologic response (e.g., peak flow timing).



Other Important Characteristics



Basin Relief

Elevation difference between highest and lowest points.



Average Slope

Affects flow velocity and erosion potential.



Climate & Precipitation

Determine the amount of water available for runoff.



Land Cover / Soil

Influences infiltration, runoff, and water storage.



Geology

Controls permeability, groundwater flow, and baseflow.



Hydrologic Response

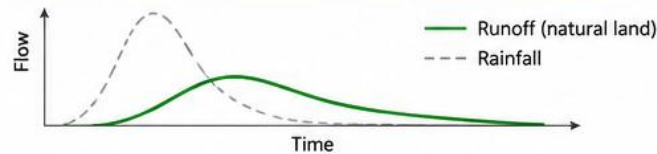
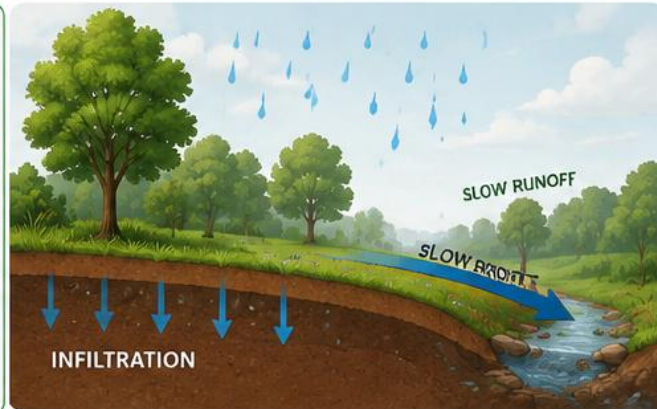
Time to peak, flood magnitude, and flow duration.

Land Cover and Imperviousness

How development changes runoff

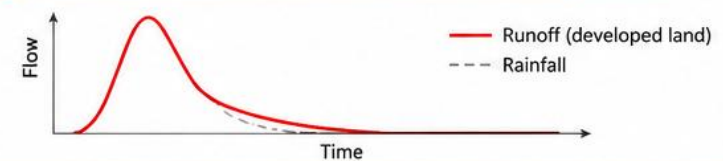
NATURAL LAND (forest, pasture)

-  **High infiltration**
Rain soaks into the soil
-  **Slow, attenuated runoff**
Lower, delayed peaks
-  **Runoff coefficient**
~0.1 – 0.3
(10–30% of rain becomes runoff)

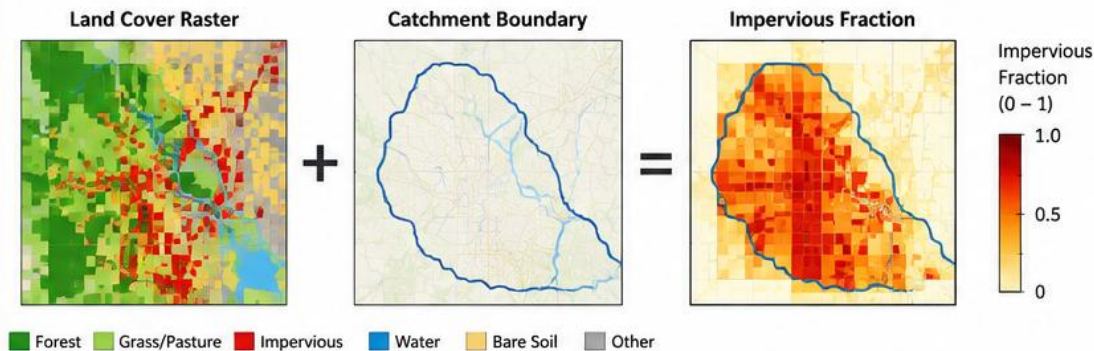


DEVELOPED LAND (roads, roofs, concrete)

-  **Low infiltration**
Rain can't soak into the soil
-  **Fast, intense runoff**
Higher, sharper peaks
-  **Runoff coefficient**
~0.7 – 0.95
(70–95% of rain becomes runoff)

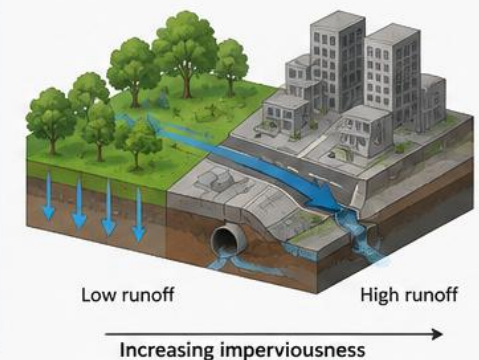


GIS OVERLAY: LAND COVER RASTER + CATCHMENT BOUNDARY = IMPERVIOUS FRACTION



Why it matters

-  **Drives stormwater design**
More impervious area = more runoff volume and higher peak flows
-  **Improves flood modelling**
Better inputs for more accurate hydrologic models
-  **Supports planning & compliance**
Used for consent requirements and regulatory reporting



Key idea: As imperviousness increases, infiltration decreases and more rain becomes quick surface runoff. GIS helps measure it, so we can design better and reduce flood risk.

Combining Hydrology with Land Use

Overlay analysis — the integration step:

Catchment boundary (from DEM) + Land use (from satellite/council data)

= Runoff characteristics for each subcatchment

Practical applications:

Stormwater modelling: estimate peak flows from subcatchment properties

Water quality: which land uses contribute pollutants to which streams?

Development impact: how does rezoning change runoff in this catchment?

Ecological assessment: how much native habitat remains in each catchment?

This is where vector analysis (land use polygons) meets raster analysis (DEM-based hydrology).

Stormwater and Flood Management with GIS

GIS enables the complete stormwater/flood assessment pipeline:

1. Delineate catchments from DEM
2. Quantify land use and imperviousness per subcatchment
3. Estimate peak flows using the Rational Method: $Q = C \times I \times A$
 - C = runoff coefficient (from land use)
 - I = rainfall intensity (from HIRDS/NIWA)
 - A = catchment area (from DEM)
4. Map flood extents: where does `water_level > ground_level`?
5. Assess exposure: overlay flood extent with buildings, roads, infrastructure

Every parameter in $Q = CIA$ comes from GIS analysis.

Climate Change Adaptation

Changing rainfall, changing catchment response:

More intense rainfall events — higher peak flows for the same catchment

Changed seasonal patterns — drought vs flood extremes

Sea level rise — coastal catchments face backwater effects

Land use change — development increases imperviousness over time

GIS enables scenario modelling:

Same catchment + different rainfall = future flood extent

Same rainfall + different land use = impact of development

Combine both = compound risk assessment

Canterbury: MfE climate projections suggest 10-20% more intense design rainfall by 2090.

KEY CONCEPTS

Today's Takeaways

1. DEMs must be preprocessed (sinks filled) before hydrological analysis
2. D8 routes each cell's water to the steepest downhill neighbour
3. Flow accumulation counts upstream cells — high values = streams
4. Catchment delineation traces all cells draining to a pour point
5. Land cover overlay converts catchment hydrology to runoff estimates

NEXT LECTURE

Lecture 9: Raster Analysis

"A catchment boundary runs through a gully — what went wrong?"

Think about the workflow:

Could there be unfilled sinks in the DEM?

Was the pour point snapped to the stream network?

Is the DEM resolution fine enough to resolve the gully?

Could the CRS be wrong (lat/lon instead of metres)?

A catchment boundary should follow ridges, never gullies — if it crosses a valley, the flow routing is broken.

Elevation Transects Along Flow Paths

Extract elevation profiles along stream channels:

1. Extract the stream network from flow accumulation
2. Trace a flow path from source to outlet
3. Sample DEM elevation along the path
4. Plot: distance (x) vs elevation (y) = longitudinal profile

Applications:

Identify knickpoints (sudden drops) — waterfalls, erosion points

Compute stream gradient for hydraulic design

Compare natural vs modified channels

A longitudinal profile tells the story of how a river drops from mountains to sea.

Common Errors in DEM-Based Hydrology

Unfilled sinks = impossible drainage patterns:

Water pools in artefact depressions and never reaches the outlet

Disconnected stream networks — 'streams' that start and stop randomly

Pour point not snapped to network:

Off by 1 cell = completely wrong catchment (or a tiny catchment)

Wrong CRS or resolution mismatch:

DEM in lat/lon (degrees) instead of metres — slope calculations are wrong

Mixing 1m DEM with 30m DEM — misaligned flow paths

Prevention: always fill sinks, always snap pour points, always check CRS.